

Name: NIKHIL RAJ

Enrollment No: 22321022

BRANCH: BSMS Chemical Science

PROJECT: Satellite Imagery-Based Property Valuation

I am doing the project with proper reading of the Project objective and instructions. In this project, I am predicting residential property prices by combining traditional housing attributes with visual environmental context derived from satellite imagery.

The dataset given is train(1).xlsx , test2.xlsx

I will analyze the train(1).xlsx dataset and do Exploratory data analysis on this.

Number of rows: 16209

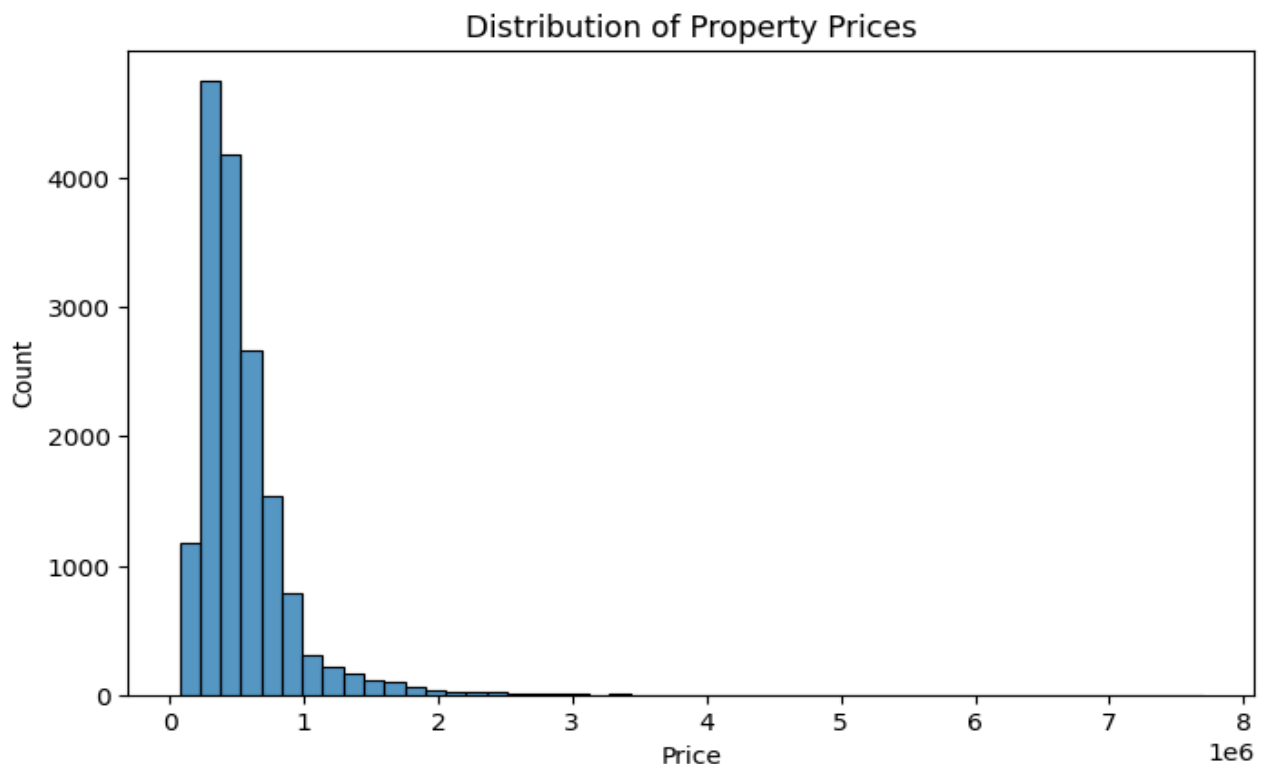
Number of columns(features) : 21

The features are: ['id', 'date', 'price', 'bedrooms', 'bathrooms', 'sqft_living', 'sqft_lot', 'floors', 'waterfront', 'view', 'condition', 'grade', 'sqft_above', 'sqft_basement', 'yr_built', 'yr_renovated', 'zipcode', 'lat', 'long', 'sqft_living15', 'sqft_lot15'].

ANALYZING THE DATASET(EDA):

The distribution of property prices is highly right-skewed, with most properties priced below one million and a small number of high-value outliers.

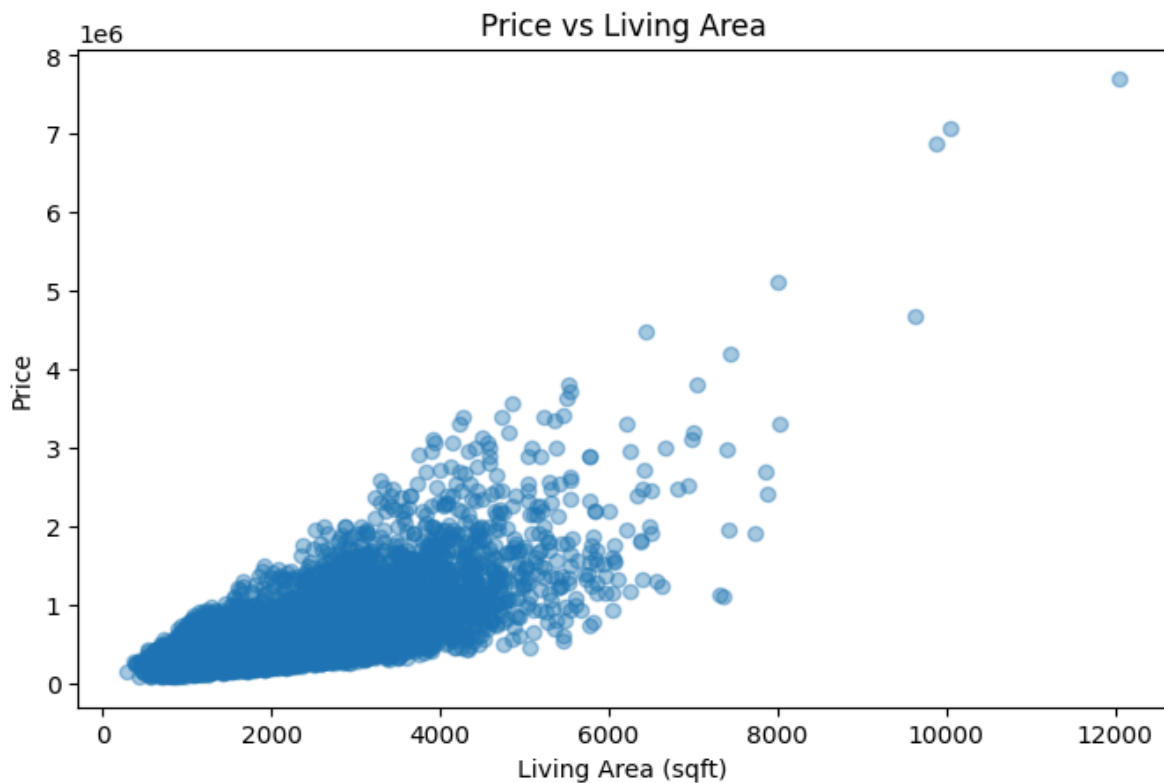
Price Distribution Histogram



Structural Features and Price:

Price vs Living Area:

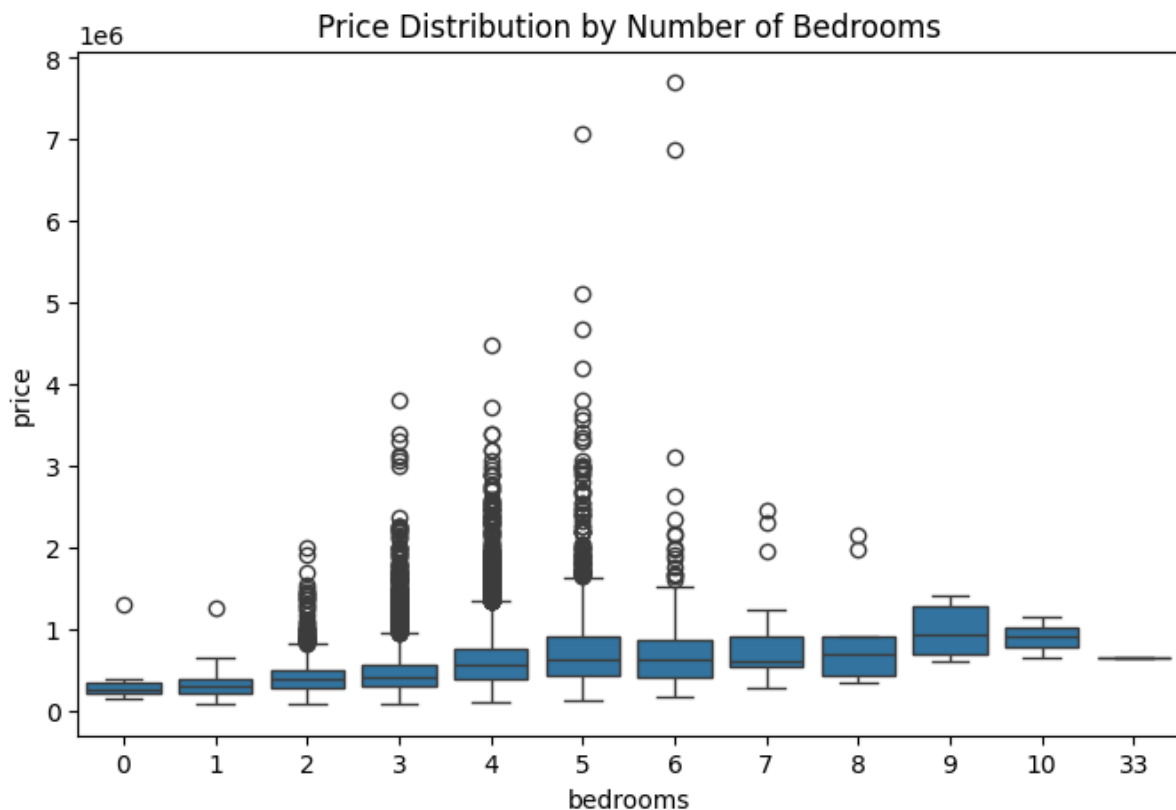
Relationship between living area and property price. The scatter plot shows a strong positive trend, indicating that larger living areas are generally associated with higher property prices.



However, the wide spread of prices for similar living areas suggests that size alone does not fully determine value, highlighting the importance of additional factors such as neighborhood characteristics and visual context.

Price vs Number of Bedrooms:

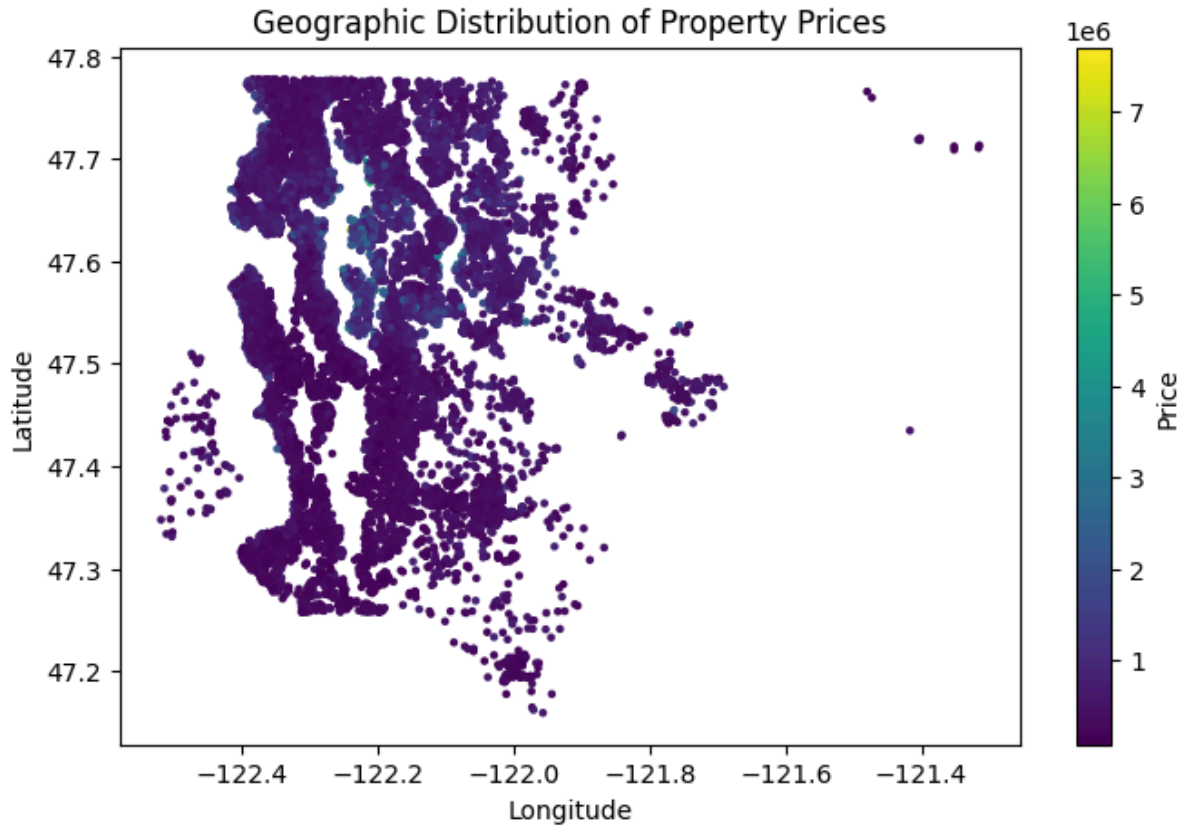
Distribution of property prices across different bedroom counts. Median prices generally increase with the number of bedrooms, indicating a positive relationship between bedroom count and property value.



However, substantial overlap and the presence of high-price outliers across multiple bedroom categories suggest that bedroom count alone does not fully explain price variation, emphasizing the role of other factors such as location and neighborhood quality.

Geographic Scatter (Latitude–Longitude):

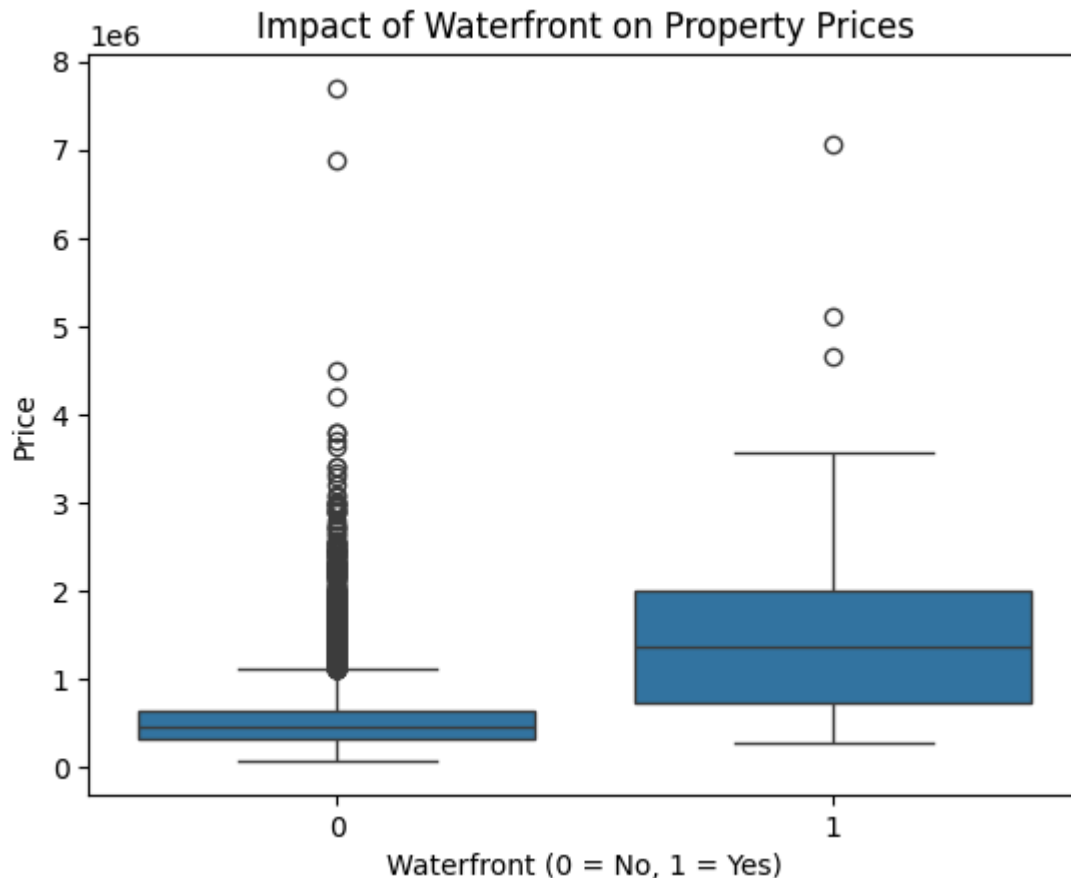
Geographic distribution of properties colored by price. The plot reveals clear spatial clustering of high-value properties in specific regions, while lower-priced properties are more widely distributed.



This spatial variation highlights the strong influence of location and neighborhood characteristics on property values and motivates the incorporation of satellite imagery to capture localized environmental context.

Waterfront vs property prices:

Properties with waterfront access exhibit significantly higher prices, reinforcing the importance of visual and environmental factors.



Finally i did the analyzation, Exploratory and geospatial analysis revealed strong spatial clustering of property prices, indicating the influence of neighborhood characteristics. Satellite imagery was used as a proxy to capture visual factors such as green cover, road density, and proximity to water, which are not directly available in tabular form.

Financial/Visual Insights : Analysis of which visual features (e.g., trees vs. concrete) drive value.

Why visual features matter financially:

Traditional housing attributes such as size and bedroom count explain a substantial portion of price variation; however, they fail to capture neighborhood aesthetics and environmental quality. Satellite imagery provides a visual representation of such factors, enabling the model to incorporate curb appeal and surrounding infrastructure into property valuation.

Green cover vs concrete

Properties surrounded by green spaces tend to exhibit higher market values compared to those located in highly concrete-dense environments. Green cover often reflects better environmental quality, reduced noise, and enhanced visual appeal, which are positively perceived by buyers. In contrast, densely built-up areas with limited greenery are generally associated with lower prices.

Road density and urban layout

Satellite images reveal variation in road density and urban layout across different neighborhoods. Well-connected areas with organized road networks often command higher prices due to improved accessibility. Conversely, overly congested regions with dense road networks may negatively impact valuation due to noise and traffic concerns.

Spatial clustering & proximity effects

The geospatial distribution of prices demonstrates clear clustering of high-value properties in specific regions, indicating strong location-based effects. These clusters often coincide with visually appealing neighborhoods, access to open spaces, or proximity to water bodies, further reinforcing the financial importance of visual environmental features.

ML interpretation (CNN + images)

By extracting CNN-based visual embeddings from satellite images, the multimodal model learns complex patterns associated with neighborhood quality, such as greenery, building density, and spatial organization. These learned

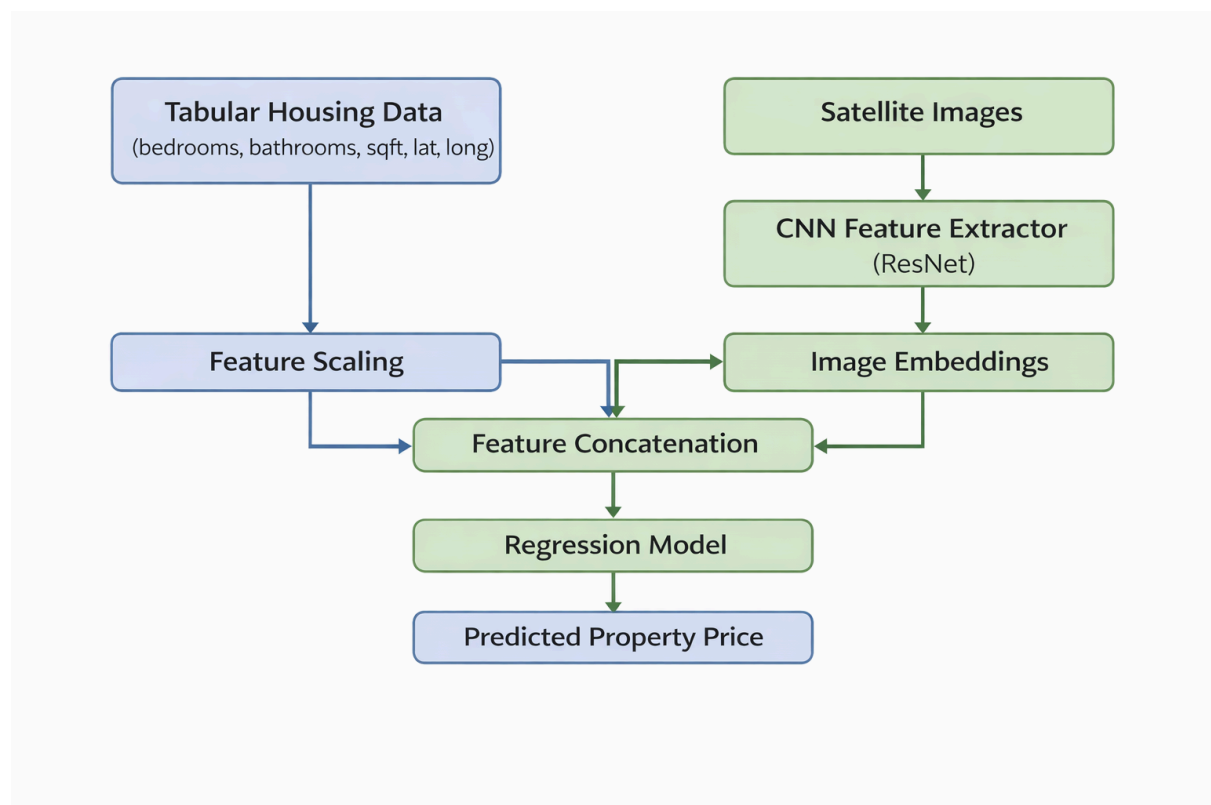
representations complement traditional tabular features, allowing the model to capture valuation drivers that are otherwise difficult to quantify.

Architecture Diagram

Architecture of the proposed multimodal property valuation framework.

Tabular housing attributes such as bedrooms, bathrooms, living area, and geographic coordinates are preprocessed and scaled, while satellite images are passed through a CNN-based feature extractor (ResNet) to obtain visual embeddings. These image features are concatenated with tabular features and fed into a regression model to predict property prices

Architecture Diagram shown below:



FEATURE SELECTION:

Now doing the Feature engineering and keeping one the key features and removing other features. Key features are

price (Target), bedrooms, bathrooms, sqft_living, lat (Latitude), long (Longitude).

Steps I done on the Google colab:

(here Tabular Data only)

Separating train dataset into X and Y/target

Doing Train Test Split

Feature Scaling

Training XGBoost Regressor

Evaluate the Model (RMSE & R^2)

RMSE: 166927.7109170314

R2: 0.7779490947723389

To improve predictive performance, the target variable was log-transformed to reduce skewness in property prices. An XGBoost regression model was trained and evaluated using 5-fold cross-validation. This approach improved the R^2 score by stabilizing variance and enhancing the model's ability to capture nonlinear relationships.

(APPLYING FOR BETTER R2 SCORE)

Log-transforming the target variable for increasing R2 Score

Train–Validation Split

Feature Scaling

Training XGBoost on log(price)

Predicting & Converting back to REAL prices

Evaluating RMSE & R^2

RMSE (log target): 160859.81492315177

R2 (log target): 0.7937989751013171

After Cross-validation

**Cross-validated R2 scores: [0.86494625 0.84996015
0.86075938 0.85356517 0.8642121]**

Mean Cross-Validated R2: 0.8586886127343192

(Now Tabular + Satellite)

I am following Multimodal Pipeline :

1)Lat / Long

2)Satellite Images

3)CNN (ResNet)

4)Image Embeddings

5)Tabular Features

6)Scaling

7)Image Embeddings + Tabular Features

8)Feature Concatenation

9)XGBoost Regressor

10)Price Prediction

SATELLITE IMAGES FETCHING PROCESS:

Sample satellite images acquired using latitude and longitude coordinates, illustrating variations in neighborhood layout, road density, and surrounding land cover across different properties.

Satellite imagery was programmatically acquired using a publicly available ESRI World Imagery tile service based on latitude and longitude coordinates, avoiding the use of paid APIs

Satellite imagery was used as a proxy to capture visual factors such as green cover, road density, and proximity to water, which are not directly available in tabular form.

I got a specific image by using index which is below:



Below sample satellite image corresponding to a residential property location. The image captures neighborhood-level characteristics such as green cover, road networks, and building density, which serve as proxies for curb appeal and environmental quality. These visual features are used by the model to complement traditional housing attributes.

Displays 6 satellite images at once



Satellite images were programmatically downloaded for all properties in both training and test datasets using latitude and longitude coordinates. To ensure robustness, image acquisition was performed in batches with error handling and resumable execution. These images were then processed using a pretrained CNN to extract visual embeddings for multimodal modeling.

-> NOW Feature Extraction from the satellite Images downloaded.

-> Then fused Tabular features + satellite features

-> Then trained the model using XG Boost.

-> Got RMSE and R2 Score

POINTS I WANT TO SHOW TO SHOW WHAT I DONE:

Visual embeddings extracted from satellite imagery using a pretrained CNN were concatenated with scaled tabular housing attributes to form a unified multimodal feature representation. This fused feature set was used to train an XGBoost regression model for property price prediction.

The multimodal model outperformed the tabular-only baseline, demonstrating that satellite imagery provides complementary neighborhood-level information that improves property valuation accuracy.

Multimodal RMSE: 160298.4071119261

Multimodal R2: 0.7952357647355146

(AFTER doing tabular + satellite images the result i got is above)

The multimodal model was evaluated using 5-fold cross-validation with R^2 as the evaluation metric. This approach provides a robust estimate of generalization

performance by averaging results across multiple train–validation splits.

*Multimodal Cross-Validated R^2 scores: [0.85936042
0.84306821 0.85263138 0.84895993 0.85615442]*

Mean Cross-Validated R^2 : 0.8520348695843939

PERFORMANCE comparison i want to conclude is that the both performed well Tabular only and Tabular + satellite Images. As i have showed the RMSE and R^2 score for both the process.

I made the final property price predictions for the test dataset were generated using the trained multimodal XGBoost model and exported in CSV format with

***property identifiers and
predicted prices, as required.***

***THANK YOU FOR
GIVING SUCH A
GOOD Project.***